

Data Complexity and Meta-Learning on Obesity Dataset

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ABSTRACT

This report will delve into the data complexity of the Obesity Levels dataset, its preprocess, and run different Data Complexity and Meta-learning packages to better understand how to take full potential of the data. Afterwards, it will be analysed the impact such metrics have on classifiers with different paradigms: Support Vector Machines, Decision Trees and K-Nearest Neighbours.

1 INTRODUCTION

Obesity is currently considered by the World Health Organization (WHO) as a global epidemic, affecting millions worldwide and posing severe health risks such as cardiovascular disease, diabetes, and certain cancers[2]. As data increases by the day in the healthcare sector, machine learning models offer the potential to utilize this data to build predictive models that identify individuals at risk of obesity.

In this report, we aim to explore these data complexities by examining the "**Obesity Levels**" dataset[5]. Additionally, we will evaluate the impact of multiple data complexity metrics and meta-learning insights on model performance, to help us understand what issues can be further improved in the dataset. In the end, we want to compare the pre-processed dataset with a BMI feature, and another one with the same features, but without this key feature.

It will be tested and compared the performance of classifiers with different paradigms: **Support Vector Machines** (SVM)[3], **Decision Trees**[10], and **K-Nearest Neighbours** (KNN)[4]. These algorithms, each with distinct approaches to data classification, allow us to explore how different learning methodologies respond to the inherent complexities of the dataset. By analyzing the results, we aim to uncover the strengths and limitations of each classifier in addressing obesity prediction.

2 METHODOLOGY

In this section, we will delve in the methodology used to produce the results obtained in the Section 3. We will pass through all the stages of decision: from dataset selection, its preprocessing, and finally tools selected to investigate these

2.1 Dataset Selection

The dataset used in this study includes data from individuals in Mexico, Peru, and Colombia, focusing on attributes related to eating habits and physical conditions for obesity level estimation. Moreover, the 2.111 records are a combination of both artificial and real samples from user collected through a web platform. It was chosen to develop this report on this topic because, even though BMI is currently the most widely used measure to quickly determine if someone is obese[6], it would be interesting to know how relevant

these other features contribute to a person's classification of their weight. Despite it being expected to BMI still outperform these other metrics (since its formula includes weight and height), the main purpose is to understand if these other features/questions are that lacking comparing to BMI. Furthermore, it will be good to investigate how the complexity of the dataset increases without this measure.

2.2 Data Pre-Processing

In this subsection, it will be exposed the preprocessing data pipeline utilized for the treatment of this dataset. It is important to highlight this topic, since it is necessary to optimize the learning models utilized in the end of the report, as well as to run some of the features of the packages explored in Subsection 2.3.

Feature Selection

In this stage we decided to select which features were the most relevant ones for the algorithm, and which ones we should drop. After careful consideration we decided to drop SCC (if the user counted their daily calorie intake), SMOKE (if the user smoked), MTRANS (which mean of transport the user used) and CAEC (if the user eats between meals).

Encoding Categorical Variables

Here we decided to encode categorical variables, in order to have all attributes in the dataset as of the numerical type. For that, we decided to use Label Encoding, since from all kept variable, all of them were ordinary, so the Label Encoding will perform well on these categorical features.

Feature Engineering

In this stage we created the feature BMI, since, as it was explained previously, is one (if not the best) measure to determine the obesity level of an individual.

Class Merging for Target Variable

The target class has a lot of classes to classify obesity, and a lot of them are just specifications of umbrella terms (for example, Overweight_Level_I and Overweight_Level_II belong to the class Overweight). So, we decided to merge those terms (except the Obesity_Type_III, since these values could be interesting to see if the results differ for the most extreme cases).

2.3 Complexity and Meta-learning Packages

For this project we used 3 different packages to understand the complexity of our dataset and to learn about the meta-features as well:

Problexty[7]

This package offers a wide variety of complexity measures to understand the underlying complexity of the dataset being studied. These features are grouped by families, which are the following:

- **Feature-based** measures (Red): Analyse the statistical properties of individual features, such as variance or correlation.
- **Linearity** measures (Orange): Evaluate the degree to which classes can be linearly separated within the feature space.
- **Neighbourhood** measures (Yellow): Assess the density and structure of data points in local neighbourhoods, providing insight into class separability.
- **Network** measures (Lime): Use graph-based approaches to capture the structure and relationships between instances within the dataset.
- **Dimensionality** measures (Green): Examine the complexity arising from the number of features relative to instances, as well as intrinsic dimensionality.
- **Class Imbalance** measures (Blue): Quantify the imbalance in the distribution of class labels, which can impact model performance.

The colours represented in the bullet points correspond to the colours these families take when you generate a complexity graph with this package .

Pymfe[9]

This package is great for gaining insights into the meta-features of any dataset. It gives us a wide range of values, from statistical measures to more complex characteristics. Pymfe helps identify patterns in the data, such as the relationships between predictor variables and the target variable, the distribution of classes, feature entropy, and other important structural aspects of the dataset. This tool is particularly useful for understanding the inherent complexity of the data, enabling more informed decisions when selecting machine learning models.

Pyhard[8]

Finally, Pyhard is great to outline not just the complexity of the dataset, but the complexity of individual instances. This is important because it can help us understand which instances are harder to classify, and why. This can be useful for understanding the limitations of our models, and for improving them. That's why when we talk about using Pyhard, we are trying to analyse the Instance Hardness of isolated points. Furthermore, Pyhard also has a UI environment, which let's us further discover how a particular dataset behaves against algorithms (like Random Forest, Gradient Boost, etc...).

3 RESULTS AND DISCUSSION

As stated before, the final objective of this report is to compare how this dataset behaves in terms of complexity if the BMI features are not present in the dataset (basically, only the features corresponding to nutritional, physical and behavioural habits of users. So, these were the results found in terms of data complexity and model performance.

3.1 Complexity Analyses Results

Regarding the complexity of the two variations of the "Obesity Levels" Dataset, we can observe that the dataset without the BMI attribute has a higher data complexity (34.8%) than the one with the BMI attribute (17.3%).

Both datasets presented approximately the same score for the network, dimensionality and class imbalance measures. Both datasets appear to have a high complexity specifically in terms of **Density** of the data's connectivity graph, and with the **Hubs** (which suggest, correctly, that the classes in our dataset have low margins of error). However, the dataset without BMI variable have significantly higher feature-based measures, and some increase in complexity in linearity and neighbourhood measures. This is to be expected, since without the BMI variable, other features must capture the relationships BMI represented, increasing feature complexity, linearity, and neighborhood measures as the dataset compensates for the missing information.

In terms of meta-features analysis also confirm to us that the dataset with the BMI parameter has more vital information to the good performance of the dataset on learning models.

The dataset with BMI attribute has a stronger gravity (18.38 compared with 7.23), lower leaves per branch and much lower leaf homogeneity (454 compared with 576913), a much better linear separability (0.9276 compared with 0.5647) and a lower tree depth (5 compared with 10). All of this just reinforce how the BMI metric facilitates the definition of obesity.

3.2 Model Performance

Lastly, as presented in the beginning, we decided to benchmark the performances on three datasets: the original "Obesity Levels" dataset (with the pre-process steps present in Section 2.2), just with the BMI parameter, and without the BMI parameter. These were the **accuracy results**, after performing a 10-fold cross validation[1] training on the algorithms in study (SVM, KNN, and Decision Trees):

Dataset	SVM Acc	KNN Acc	Decision Tree AcC
Original	0.9285	0.8248	0.9825
Only BMI	0.9607	0.9692	0.9578
Without BMI	0.7429	0.7552	0.7458

Table 1: Model accuracy results across different datasets

The model with the best performance on the original dataset was the Decision Tree (0.9825 accuracy), likely due to its ability to capture non-linear relationships and hierarchical structures in the data. In contrast, SVM and KNN may have underperformed because they focused heavily on the BMI attribute when building their classification models, which caused the remaining features to act more like noise than useful information for classification. This also explains why SVM and KNN outperformed Decision Trees on the dataset with only the BMI attribute.

4 CONCLUSIONS

To sum up, this study shows how important BMI is to learning models when trying to predict the obesity level of an individual.

Just focusing on eating habits, physical activity and other human behaviours isn't enough to safely determine the obesity of somebody, because of two reasons: there might be underlying health issues, genetic predisposition, or just different metabolisms, which cause someone to be overweight or underweight; and people might currently follow eating habits different than what is expected for their obesity level (e.g. someone could try to gain weight). However, these health habits could be seen as a future trend for most individuals, and when aligned with BMI and the right learning model, it can be a powerful tool in determining obesity.

Moreover, the complexity analyses and meta-feature learning helped a lot in predicting how well some models might behave in comparison with other on the same data, augmenting efficiency in a data pipeline.

REFERENCES

- [1] Daniel Berrar et al. 2019. Cross-validation.
- [2] Victoria A Catenacci, James O Hill, and Holly R Wyatt. 2009. The obesity epidemic. *Clinics in chest medicine* 30, 3 (2009), 415–444.
- [3] Vinod Kumar Chauhan, Kalpana Dahiya, and Anuj Sharma. 2019. Problem formulations and solvers in linear SVM: a review. *Artificial Intelligence Review* 52 (2019), 803–855.
- [4] Gongde Guo, Hui Wang, David Bell, Yaxin Bi, and Kieran Greer. 2003. KNN model-based approach in classification. In *On The Move to Meaningful Internet Systems 2003: CoopIS, DOA, and ODBASE: OTM Confederated International Conferences, CoopIS, DOA, and ODBASE 2003, Catania, Sicily, Italy, November 3-7, 2003. Proceedings*. Springer, 986–996.
- [5] Fatemeh Mehrparvar. 2023. Obesity Levels Dataset. <https://www.kaggle.com/datasets/fatemehmehrparvar/obesity-levels>. Accessed: 2024-11-10.
- [6] Francisco B Ortega, Xuemei Sui, Carl J Lavie, and Steven N Blair. 2016. Body mass index, the most widely used but also widely criticized index: would a criterion standard measure of total body fat be a better predictor of cardiovascular disease mortality?. In *Mayo Clinic Proceedings*, Vol. 91. Elsevier, 443–455.
- [7] Proplexity Developers. 2023. *Proplexity Documentation*. <https://proplexity.readthedocs.io/en/latest/index.html>. Available at <https://proplexity.readthedocs.io/en/latest/index.html>.
- [8] PyHard Developers. 2023. *PyHard Documentation*. <https://ita-ml.gitlab.io/pyhard/index.html>. Available at <https://ita-ml.gitlab.io/pyhard/index.html>.
- [9] PyMFE Developers. 2023. *PyMFE Documentation*. <https://pymfe.readthedocs.io/en/latest/index.html>. Available at <https://pymfe.readthedocs.io/en/latest/index.html>.
- [10] J. Ross Quinlan. 1996. Learning decision tree classifiers. *ACM Computing Surveys (CSUR)* 28, 1 (1996), 71–72.